FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Main)-2021

TEST DATE: 04-01-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

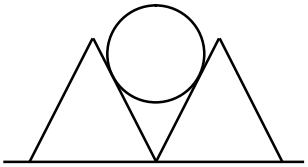
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A particle starts with 10 m/s speed from origin and travels half of the remaining distance in every 1s interval then its position after very long time can be written as
   (A) $10\ell n2$
   (B) $\frac{10}{\ell n2}$
   (C) 10
   (D) $\ell n2$

2. A water sprinkler is throwing water with same speed in every possible direction in a plane. The locus of outer tangent curve touching all parabolic trajectories will be
   (A) Circle
   (B) Parabola
   (C) Straight line
   (D) None

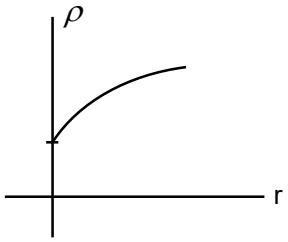
3. Select correct statement from the following -
   (A) A particle moving in elliptical path has radius of curvature equals to its semi major axis when it is at farthest distance from center.
   (B) An observer on accelerated plank must not apply pseudo force on plank itself.
   (C) A particle having constant acceleration cannot have constant radius of curvature in 2-D Motion.
   (D) Angular acceleration can only be along angular velocity.

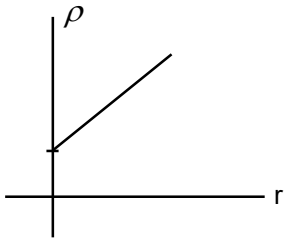
4. A smooth disc is resting on two equilateral triangles as shown in figure. All bodies have same masses. Find distance of normal force line by ground on any triangle from its center. (assume disc is touching at mid points of each triangle's side)

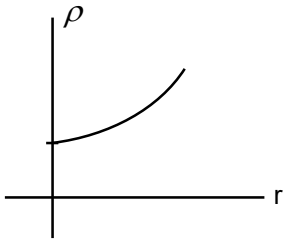


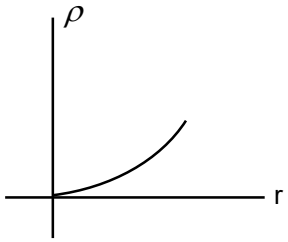
   (A) $\frac{a}{5}$
   (B) $\frac{a}{3}$
   (C) $\frac{a}{6}$
   (D) $\frac{a}{4}$

5. A compressible fluid is rotated in a vertical cylindrical container with uniform angular velocity about symmetrical axis. Due to pressure, density of fluid increases and it is observed that open surface of liquid remains horizontal. Assume density only varies radially, not vertically. Which of the following graph shows correct variation of density with radial distance at same horizontal level?

(A) 

(B) 

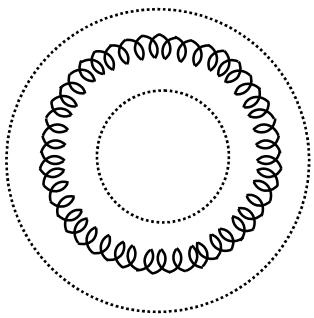
(C) 

(D) 

6. An ideal gas in closed container goes through a process such that internal energy of gas is always inversely proportional to its density then-
   (A) Pressure of gas has parabolic variation with Temperature.
   (B) Volume of gas varies linearly with Temperature.
   (C) Pressure of gas varies inversely with Temperature.
   (D) Volume of gas varies inversely with Temperature.

7. A thin elastic loop of natural length ℓ is kept slightly on a water film in gravity free space and inner film of loop is bursted keeping outer film intact. If force constant of loop is k and loop follow hook's law then radius of loop in equilibrium is (Surface Tension of water is S)
   (A) $\dfrac{k\ell}{\pi k - S}$
   (B) $\dfrac{\ell}{2\pi}$
   (C) $\dfrac{2k\ell}{\pi - S}$
   (D) $\dfrac{k\ell}{2\pi k - 2S}$

8. A spring of mass M and force constant K in form of a circular loop performing oscillations moving radially in and out in free space keeping its center at rest. Time period of oscillation is given by (Assume cross sectional radius of spring is very small than radius of spring loop)



   (A) $\sqrt{\dfrac{M}{K}}$
   (B) $2\pi\sqrt{\dfrac{M}{K}}$

(C) $2\pi\sqrt{\frac{M}{3K}}$

(D) $2\pi\sqrt{\frac{2M}{K}}$

9. In a resonance tube experiment first and second resonance are heard at ℓ_1 and ℓ_2 marks on meter scale then least percentage error in measurement of end correction is when

(A) $\ell_1 = 3.0\,cm$, $\ell_2 = 15.0\,cm$

(B) $\ell_1 = 2.0\,cm$, $\ell_2 = 8.0\,cm$

(C) $\ell_1 = 2.0\,cm$, $\ell_2 = 16.0\,cm$

(D) $\ell_1 = 1.0\,cm$, $\ell_2 = 7.0\,cm$

10. A uniformly charged semicircular ring of unit radius with linear charge density λ placed in x-y plane centered at origin in positive y region. A potential was already there in space, variable with space coordinates as $V = \left(4x^2 - 1\right)y$

Then potential energy of ring excluding self energy is

(A) $\frac{2}{3}\lambda$

(B) λ

(C) $\frac{\lambda}{2}$

(D) 2λ

11. A square plate is cut from a disc of radius R such that diagonal of square is diameter of disc. A point charge q is placed on axis of disc at $\frac{R}{\sqrt{2}}$ distance from center. Value of electric flux passing through remaining disc is

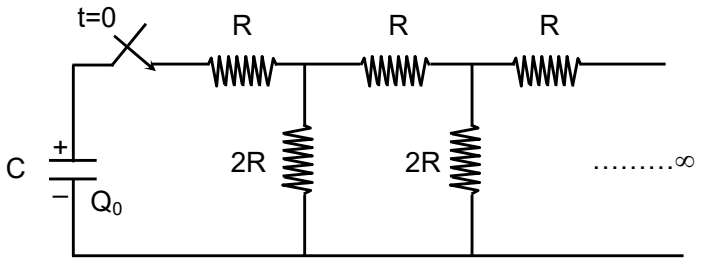
(A) $\frac{q}{6\varepsilon_0}\left(5 - \sqrt{3}\right)$

(B) $\frac{q}{12\varepsilon_0}\left(5 - 2\sqrt{3}\right)$

(C) $\frac{q}{3\varepsilon_0}\left(5 - \sqrt{3}\right)$

(D) $\frac{q}{3\varepsilon_0}\left(5 - 2\sqrt{3}\right)$

12. A capacitor is being discharged by infinite ladder of resistors, starting from Q_0 charge. Total heat dissipated in horizontal resistors is



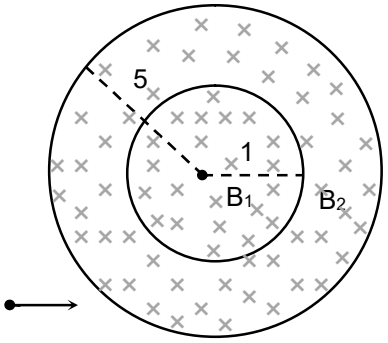
(A) $\frac{Q_0^2}{4C}$

(B) $\frac{Q_0^2}{6C}$

(C) $\frac{Q_0^2}{3C}$

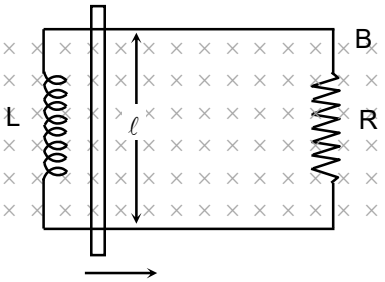
(D) $\frac{Q_0^2}{5C}$

13. A positive charge enters horizontally into a magnetic field confined in circular region in gravity free space where 1m is radius of inner circle and outer circle has 5m radius. Eventually particle enters tangentially vertically into inner circle from right most point of inner circle and starts revolving just on inner edge of circle. Find Ratio of magnetic fields in two regions B_1/B_2



(A) 2
(B) 4
(C) 5
(D) 3

14. A rod can slide on smooth conducting frame connected with inductor and resistor in a uniform magnetic field. Rod has negligible resistance and it is given a push towards positive x-axis with initial kinetic energy K. The energy lost in heat till first time rod comes to rest, after traveling a distance x, is



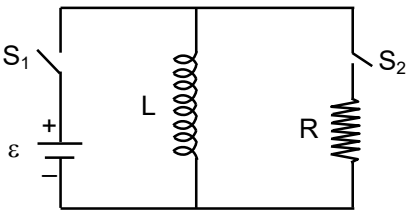
(A) $\frac{K}{2}$

(B) K

(C) $K-\frac{B^2\ell^2x^2}{L}$

(D) $K-\frac{B^2\ell^2x^2}{2L}$

15. In the circuit shown in figure initially all switches are open and at t = 0 S_1 is closed keeping S_2 open. After $t = t_0$ S_1 is opened and S_2 is closed then total energy lost in this process after long time is



(A) $\frac{\varepsilon^2t_0^2}{L}$

(B) $\frac{\varepsilon^2t_0^2}{3L}$

(C) $\frac{\varepsilon^2t_0^2}{2L}$

(D) $\frac{3\varepsilon^2t_0^2}{L}$

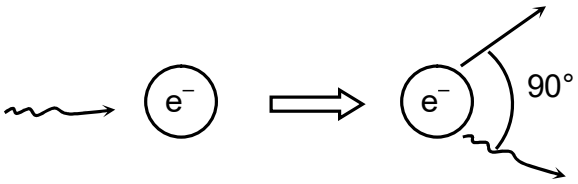
16. In displacement method a lens is moving with uniform speed 1 cm/s from fixed object towards fixed screen. It is observed that two images of twice and half size of object are sharply obtained in a time duration of 6s, then focal length of lens is
(A) 4 cm
(B) 6 cm
(C) 3 cm
(D) 2 cm

17. Only thin film shows interference because
(A) In thick film light is absorbed in large amount.
(B) In thick film interference occurs but fringes are washed out.
(C) In thick film interference is not for visible wavelengths.
(D) Light get scattered by thick film.

18. A perfectly absorbing plate is in equilibrium above sun due to gravitational pull of sun and radiation pressure. Sun is emitting P power from its surface. If density of plate is ρ. Then thickness of plate is
(Assume size of plate is very small compared to sun as it is placed perpendicular to incident radiation M_s = Mass of Sun
(A) $\frac{2P}{\pi GM_s\rho c}$
(B) $\frac{P}{\pi GM_s\rho c}$
(C) $\frac{P}{4\pi GM_s\rho c}$
(D) $\frac{P}{2\pi GM_s\rho c}$

19. A practical silicon diode does not have-
(A) Electrons in P region.
(B) Holes in N region.
(C) Capacitance of depletion layer.
(D) Zero resistance in forward biasing.

20. A photon is incident on an electron as shown after striking, electron moves at some angle and another photon of double wavelength is observed going perpendicular to electron then wavelength of incident photon is (M_e = mass of electron)

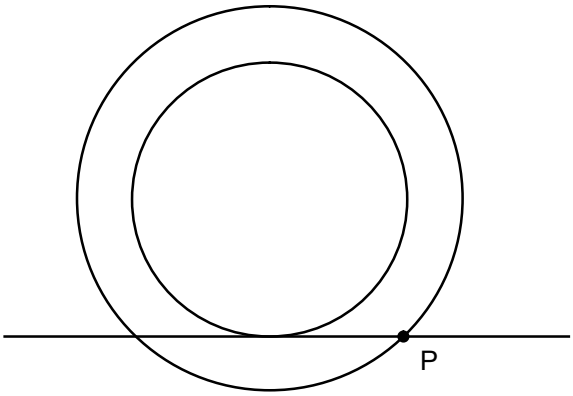


(A) $\frac{3h}{2M_ec}$
(B) $\frac{3h}{4M_ec}$
(C) $\frac{h}{2M_ec}$
(D) $\frac{2h}{M_ec}$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A spool is rolling on horizontal surface with its inner radius in contact with surface. The outer surface of spool performs cycloid motion going above and below the horizontal level. Any point P on outer circumference has velocity at 30N degrees angle with horizontal, when it is just crossing horizontal level. Then find the value of N.

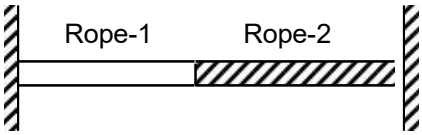


22. Moment of inertia of a rod placed along X-axis is found same about lines x = 2m & x = 8m equals 92kgm^2. Linear mass density of rod is increasing along X-axis with total mass 10 Kg. Then moment of inertia about axis passing through center of mass perpendicular to rod is?

SECTION – C
(Numerical Answer Type)

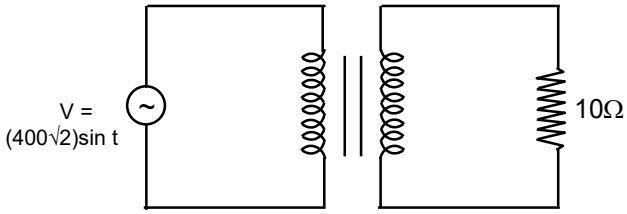
This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. Two ropes of same cross-sectional area are joined together and a sinusoidal pulse of total energy 9J is sent in first rope. Find the total energy in joule of transmitted pulse in second rope. Ratio of linear mass density of first rope to second is 4.



24. A metal ball is surrounded by a uniform resistive medium spread till infinity uniformly. Ball carries some charge and allowed to discharge through medium. Time constant of discharging is? (Permittivity of medium = 2×10^{-10} C^{2}N^{-1}m^{-2}, Resistivity of medium = 1.6×10^{10} C^{-2}Nm^{2}s)

25. A transformer with 80% efficiency is used with an AC source. Turn ratio of primary to secondary coil is 2. Find current flowing in primary coil. Assume there is no flux leakage in transformer.



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. The dissociation constant for $Ph-\underset{Cl}{\underset{|}{CH}}-COOH$ and HCN at 25° are 1.5×10^{-5} and 4.5×10^{-3} respectively; the equilibrium constant for $CN^- + Ph-\underset{Cl}{\underset{|}{CH}}-COOH \rightleftharpoons HCN + Ph-\underset{Cl}{\underset{|}{CH}}-COO^-$ would be
(A) 3×10^{2}
(B) 3×10^{-2}
(C) 3.33×10^{-3}
(D) 6.75×10^{-7}

27. False statement about proteins is
(A) Sequence of amino acids refers to primary structure
(B) α – helix pleated structures relate secondary structure
(C) 3-dimensinal folding of entire protein is tertiary structure
(D) Denaturation of protein is called annealing

28. What products are expected from the disproportionation reaction of hypochlorous acid?
(A) $HClO_3$ and Cl_2O
(B) $HClO_2$ and $HClO_4$
(C) HCl and Cl_2O
(D) HCl and $HClO_3$

29. An aqueous solution containing 28% by mass of a liquid A (molecular mass =140) has a vapour pressure of 160 mm of Hg at 37^0C.Find the vapour pressure of the pure liquid A(The vapour pressure of water at 37^0C is 150mm of Hg)
(A) 310 mm
(B) 360 mm
(C) 672 mm
(D) 72 mm

30. The correct representation of Borax is
(A) $Na_2\left[B_4O_5(OH)_4\right].8H_2O$
(B) $Na_2\left[B_4O_7\right].10H_2O$
(C) $Na_2\left[B_4O_4(OH)_4\right].9H_2O$
(D) $Na_2\left[B_4O_5(O_2)(OH)_2\right].9H_2O$

31. H_2O_2 turns blackened lead paintings to white colour in this reaction it oxidizes PbS to $PbSO_4$ the number of moles of H_2O_2 needed to oxidize 0.1 moles of PbS is
(A) 1 mole
(B) 0.1 mole
(C) 0.5 mole
(D) 0.4 mole

32. Solid PBr_5 exists as
(A) $[PBr_4]^+ [PBr_6]^-$
(B) $[PBr_4]^+ [Br]^-$
(C) $[PBr_3] [Br_2]$
(D) PBr_5

33. Acidified water was electrolysed using an inert electrode and the volume of gases liberated at STP was 168 ml. The amount of electricity passed through the acidified water was,
(A) 96,500C
(B) 9,650 C
(C) 965 C
(D) 168 C

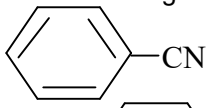
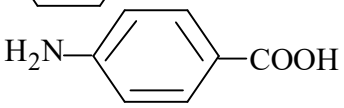
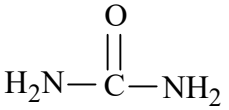
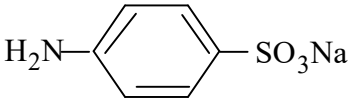
34. $RNH_2 \xrightarrow{CHCl_3/\bar{O}H}$ (X)

$R_2NH \xrightarrow{CHCl_3/\bar{O}H}$ (Y)

The major product (X) and (Y) are respectively
(A) RCN and RNC
(B) RNC and R_2NCHO
(C) both RNC
(D) RNC and No reaction

35. 250 ml of x M solution and 500 ml of y M solution of a same solute are mixed and diluted to 2 L to produce a final concentration of 1.6 M. If x : y is 5 : 4 then (x + y) is
(A) 8.06
(B) 8.86
(C) 9.8
(D) 12.6

36. The empirical formula of sheet silicate & the number of oxygens shared per tetrahedral are respectively,
(A) $\left(SiO_3\right)_n^{2n-}$,2
(B) $\left(Si_2O_7\right)_n^{6n-}$,3
(C) $\left(Si_2O_5\right)_n^{2n-}$,3
(D) $\left(Si_4O_{11}\right)_n^{6n-}$,4

37. Which of the following compounds gives blood red colour in Lassaigne's test ?
(A) 
(B) 
(C) 
(D) 

38. 5 ml of a gaseous hydrocarbon was subjected to combustion with 30 ml of O_2. The resulting gaseous mixture on cooling is found to measure 25 ml. When the gaseous mixture was passed through NaOH solution, 10 ml of the gas is absorbed and the remaining gas is absorbed by pyrogallol. All measurements are made at constant pressure and temperature. The molecular formula of the hydrocarbon is.

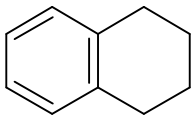
(A) C_2H_2

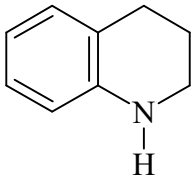
(B) C_3H_8

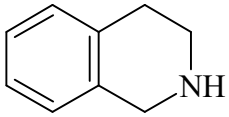
(C) C_2H_4

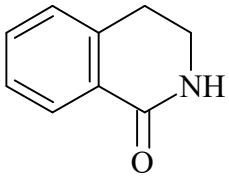
(D) C_2H_6

39. Which of the following compounds undergo bromination by electrophilic aromatic substitution reaction at the fastest rate?

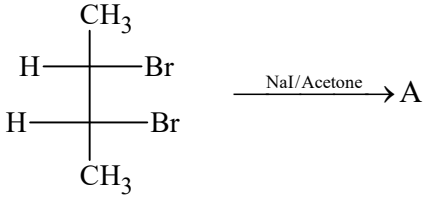
(A) 

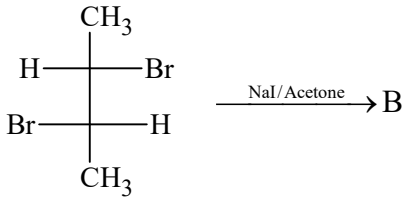
(B) 

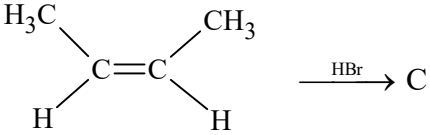
(C) 

(D) 

40. In the following reaction, which statement is true?

X 

Y 

Z 

(A) X is optical inactive, B is cis alkene, C is a single optical active compound

(B) B is cis alkene, Y is optical active, C is a single optical active compound. A is cis alkene too.
(C) A is trans alkene, B is cis alkene, C is mixture of optical active compounds. X is optical inactive compound.
(D) Y is optcal active, B is trans alkene, C is mixture of optical active compound, X is optical inactive compound.

41. The s-block hydride with covalent polymeric structure are
(A) LiH
(B) BeH_2
(C) NaH
(D) CaH_2.

42. A steel vessel of volume one litre is filled with a mixture of CH_4 and O_2 in 1: 2 mole ratio at a total pressure of 1 atm at 27^0C. The gas mixture is exploded for complete reaction. What would be the final pressure of the gaseous product at 127^0C.
(A) 3.13 atm
(B) 1.33 atm
(C) 1.13 atm
(D) 2.08 atm

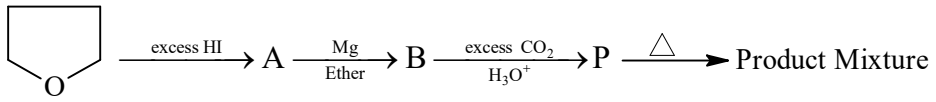
43. What is the emf of a cell containing two hydrogen electrodes, the negative one in contact with 10^{-6} M OH^- and the positive one in contact with a 0.1 M CH_3COONH_4 solution? Given pKa (CH_3COOH) = 4.74 and pKa (NH_4^+) = 9.26
(A) 0.059 volt
(B) 0.59 votl
(C) 0.0059 volt
(D) 5.9 volt

44. Phenol [C_6H_5OH] and water are taken in the weight ratio x : y in a stoppered flask and shaken thoroughly. After system settles down, it is observed that two layers are formed, one of which contains 70% phenol and the other 10% phenol. If the layers are equal in weight what is x : y ?
(A) 2 : 3
(B) 1 : 2
(C) 3 : 5
(D) 3 : 4

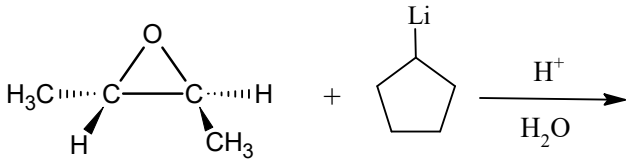
45. Amongst the following statements
I : The structure of $[BrF_4]^+$ is regular tetrahedral
II : Bond order of O_2 decreases by the removal of an electron
III : Br_3^- and Br_3^+ are having same shape and same structure
IV : The hybrid orbital of phosphorous in PCl_5 is sp^3d_{xy}
The incorrect statements is/are
(A) only II
(B) I, III, IV
(C) I, II, III
(D) I, II, III, IV

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. 

The number of carbon atoms present in the organic product which is formed in the product mixture as per the given synthesis?

47. 

The number of stereocentres in the final product is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. Given $Pt\,|\,H_2\,(g, 0.1atm)\,|\,H^+\,(aq, 10^{-2}\ M)\,||\,MnO_4^-\,(aq, 0.1\,M)\,|\,Mn^{2+}\,(aq, 0.01\,M)\,|\,Pt$

The potential of the above cell will be in volt is:

$E^{\circ}_{MnO_4^-|Mn^{2+}} = 1.5$ V

49. Ratio of number of antibonding molecular orbital electrons to the number of bonding molecular orbital electrons of O_2 is

50. In a crystalline solid, having formula AB_2O_4, oxide ions are arranged in cubic close packed lattice while cations A are present in tetrahedral voids and cations B are present in octahedral voids. The percentage of tetrahedral voids occupied by A is:

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. Let $\alpha^3 - 2\alpha^2 - 2\alpha + 3 = \beta^3 - 2\beta^2 - 2\beta + 3 = 0$ such that $\alpha \neq \beta$; $\alpha, \beta \neq 1$. Then equation having roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is
(A) $3x^2 + 4x + 3 = 0$
(B) $3x^2 + 5x + 3 = 0$
(C) $3x^2 + 7x + 3 = 0$
(D) none of these

52. The equations $x^3 + 3x^2 - 3x - 5 = 0$ and $x^3 + ax^2 + bx = 0$ have a common root, then a – b (a, b, $\in$ Q) cannot be equal to
(A) 1
(B) 5
(C) 7
(D) none of these

53. If number of rational terms in the expansion of $\left(\sqrt{2}+\sqrt{3}+\sqrt{6}\right)^{2020}$ and $\left(\sqrt{2}+\sqrt{3}+\sqrt{6}\right)^{2021}$ be p and q respectively, then p – q equals
(A) 1008
(B) 1001
(C) 1010
(D) 1011

54. Let $f(x) = \begin{vmatrix} \cos(x+\alpha)+i\sin(x+\alpha) & \cos(x+\beta)+i\sin(x+\beta) & \cos(x+\gamma)+i\sin(x+\gamma) \\ \sin(x+\alpha)-i\cos(x+\alpha) & \sin(x+\beta)-i\cos(x+\beta) & \sin(x+\gamma)-i\cos(x+\gamma) \\ \sin(2\beta-2\gamma) & \sin(2\gamma-2\alpha) & \sin(2\alpha-2\beta) \end{vmatrix}$, then the value of $f(\alpha) + f(\beta) - 2f(\gamma)$ is dependent on
(A) α
(B) β
(C) γ
(D) none of these

55. Let A, B be two matrices such that AB = A and BA = B, then $(A + B)^{2020}$ equals
(A) 2020 I
(B) 2^{2020} I
(C) 2^{2019} (A + B)
(D) none of these

56. In a game show, there are three locked chests A, B and C. One of them holds a prize and other two are empty. In a turn, contestant chooses one of the chests and flips a fair coin. If the result is heads, he gets the contents of the chest else not. Given that chest A, B and C may contain the prize with probability of $\frac{8}{13}, \frac{4}{13}$ and $\frac{1}{13}$ respectively and the contestant is given 2021 turns, then number of times he must choose chest C to maximize his chances of obtaining the prize is

(A) 0
(B) 155
(C) 672
(D) none of these

57. Area bounded by the curves $y = \sin^{-1}(\sin x)$ and $y = \cos^{-1}(\cos x)$ between x = -2π and x = 2π equals
(A) π^2
(B) $2\pi^2$
(C) $4\pi^2$
(D) none of these

58. A particle travels on the curve $y = x^3$ starting from (0, 0). If it travels at a uniform speed of 1 units/s, then at what time will its speed along y-axis will be atleast 10 times its speed along x-axis?
(A) 4
(B) 5
(C) 6
(D) none of these

59. Let 'f' be a polynomial function of degree 'n' such that $f''(x) \neq 0$ at any of the critical points, then which of the following must be true
(A) if 'n' is odd, then 'f' must have equal number of points of local maxima and local minima
(B) if 'n' is even, then 'f' must have equal number of points of local maxima and local minima
(C) 'f' has atleast $\frac{n}{2}$ points of extrema if 'n' is even
(D) 'f' has atleast $\frac{n-1}{2}$ points of extrema if 'n' is odd

60. Let f(x) be a continuous function such that $\int_0^n x f'(x)dx + \sum_{r=1}^{n} f(r) = 0$, then $\int_0^n \{x\} f'(x)dx$ equal ($n \in$ N) (where {.} denotes the fractional part of function)
(A) –nf(n)
(B) –(n + 1)f(n + 1)
(C) –(n + 1)f(n)
(D) none of these

61. Equation of curve which passes through point (1, 1) and satisfies the differential equation $3xy^2\,dy = (x^2 + 2y^3)dx$ is
(A) $y = \ln(x\,e)$
(B) $y^6 = x^4 \ln(x^2\,e)$
(C) $y^3 = x^2 \ln(ex)$
(D) $y^3 = x \ln(ex^2)$

62. Let $\bar{x}, \bar{y}$ be non-zero vectors which are equally inclined to vector $\bar{z}$ such that $\bar{x} \cdot (\bar{y} \times \bar{z}) = 0$, then $\bar{z}$ equals
(A) $\frac{|\bar{x}|\bar{x} + |\bar{y}|\bar{y}}{|\bar{x}|^2 + |\bar{y}|^2}$
(B) $\frac{|\bar{x}||\bar{y}|^2\bar{x} + |\bar{x}|^2|\bar{y}|\bar{y}}{|\bar{x}|^2 + |\bar{y}|^2}$

(C) $\dfrac{|\bar{x}||\bar{y}|(\bar{x}+\bar{y})}{|\bar{x}|^2+|\bar{y}|^2}$

(D) none of these

63. Number of solutions for the system of equations represented by
$2x + y – z – 1 = 0$
$2y + 4x – 2z = 5$
$3z – 3y – 6x = 3$ is
(A) 0
(B) 1
(C) infinite solutions all of which lie on a line
(D) infinite solutions all of which lie on a plane

64. Shortest distance between the curves $y^2 = 4x$ and $(x – 4)^2 + y^2 = 1$ equals
(A) $2\sqrt{3}-1$
(B) $2\sqrt{2}-1$
(C) $2\sqrt{2}$
(D) $\sqrt{2}+1$

65. Maximum value of area of triangle that can be inscribed in $\dfrac{x^2}{16}+\dfrac{y^3}{3}=1$ is
(A) $3\sqrt{3}$
(B) $\dfrac{9}{2}$
(C) 9
(D) $9\sqrt{3}$

66. If shortest distance between two tangents to the curve $\dfrac{x^2}{2}-y^2=1$ is 2, then area of quadrilateral made by the tangents' points of intersection with coordinate axes equals
(A) $3\sqrt{2}$
(B) $6\sqrt{2}$
(C) $\sqrt{12}$
(D) $\sqrt{24}$

67. Let $\alpha, \beta, \gamma, \delta$ be 4 distinct solutions of the equation $\tan 4x = 1$, then harmonic mean of $\tan\alpha$, $\tan\beta$, $\tan\gamma$ and $\tan\delta$ equals
(A) 1
(B) 16
(C) 4
(D) none of these

68. There is a circular path with centre at O. A and B are two diametrically opposite points on its circumference. On the line AOB, lies a point P on which a pole is erected. If pole subtends angle of 45° and 30° at A and B respectively, then angle subtended by OP at top of pole equals
(A) $\tan^{-1}\left(\dfrac{\sqrt{3}+1}{4}\right)$

(B) $\tan^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$

(C) $\tan^{-1}\left(\frac{2}{\sqrt{3}+1}\right)$

(D) none of these

69. The logical statement $(p \leftrightarrow q) \leftrightarrow (q \leftrightarrow r)$ is equivalent to
(A) $p \leftrightarrow r$
(B) $r \leftrightarrow q$
(C) $p \leftrightarrow q$
(D) none of these

70. Median of a set of 100 observations is 100. If 51 observations are increased by 5, then median of resultant data set is
(A) 95
(B) 100
(C) 105
(D) cannot be determined with given information

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. Given that $\lim_{n\to\infty} \frac{1}{Q(n)} \sum_{r=1}^{n} \frac{1}{\sqrt{a_r} + \sqrt{a_{r+1}}} = 0$ where $a_1, a_2, ...$, a_{n+1} are in A.P., a_1 is a constant and $\frac{1}{Q(n)}$ is a polynomial in 'n' having degree 'p', then lowest value of 'p' equals

72. Let A, B be points on the circle $x^2 + y^2 = 1$ such that AB = 1. Let X be locus of a point P such that area of ΔPAB equals $\frac{1}{4}$. Let Y be locus of point Q such that tangents from Q to the circle are perpendicular, then $n(X \cap Y)$ equals

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Number of values of 'z' satisfying $z^{2021} - i = 0$ $\left(i = \sqrt{-1}\right)$ such that Re(z) > 0 equals

74. Number of onto functions 'f' that can be defined from set A = {1, 2, 3, 4, 5, 6} to set A such that $f(i) \neq i$, f(1) = 2 and f(3) = 4 equals

75. Given three points A(1, 2), B(5, 7) and C(6, –2). Let P be any point on line segment AC from which a line is drawn, perpendicular to BC, intersecting line AB at D. If from point P, line be drawn perpendicular to CD, intersecting AB at (x, y), then x + y equals